Chapter 26 Problem Set

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1. If B has a minimum element then α is rational. Otherwise, it's irrational.

2. Assume there does exist a smallest positive real number, which we can call α. We know that α is less than or equal to all positive rational numbers, so if we wrote it as a Dedekind cut, (A, B), B is the set of all positive rationals, which means the maximum element of A is 0, since it's less than α and it's rational. But A can't have a maximum element, so such an α doesn't exist.

3. Let $\alpha = (A_\alpha, B_\alpha)$ and $\beta = (A_\beta, B_\beta)$. Since $\alpha < \beta$, $A_\alpha \subseteq A_\beta$. This means there exists $r \in A_\beta$ such that $r \notin A_\alpha$. By the definition of a Dedekind cut, r is rational, less than β, and greater than α.

4. Given α and β, we know that we can find $c \in \mathbb{Q}$ such that $\alpha < c < \beta$. We can also find $d \in \mathbb{Q}$ such that $c < d < \beta$. Now, let $s = c + \frac{d-c}{\sqrt{2}}$. Since $d > c$, $s > c$. Also, since $\frac{d-c}{\sqrt{2}} < d - c$, we know that $s < d$. We have found a number between c and d, and therefore α and β, so we are left to prove that s is irrational.

We see that s is irrational if and only if $\frac{d-c}{\sqrt{2}}$ is irrational. Let $x = d - c$. This gives us $\frac{x}{\sqrt{2}} = \frac{x}{2}\sqrt{2}$. Since $\frac{x}{2}$ is rational, the product of $\frac{x}{2}$ and $\sqrt{2}$ must be irrational, so s is irrational.

5. Let the countable set be $\mathbb{Q}$. Then let a subset contain all the rationals between $-\infty$ and some real number α. If $\alpha = (A, B)$ which is a Dedekind cut, then the subset is essentially just A. If we were to do this for all possible values of α, we would get an uncountable number of subsets. They are nested because, if we choose any two different subsets A_α and A_β, if $\alpha < \beta$ then $A_\alpha \subseteq A_\beta$, and $A_\beta \subseteq A_\alpha$ otherwise.

6. I couldn't find out what was wrong with this definition.

7. Let A be the set of rational numbers less than $x = \sqrt{2}$ and B the set of rationals greater than x. To show that (A, B) is a Dedekind cut, we need to verify the conditions:

A and B are nonempty: Since 1 is rational, $1 \times 1 = \langle 1, 1\rangle \times \langle 1, 1\rangle = \langle 1, 1\rangle = 1$. Similarly, $2 \times 2 = 4$. Since 2 is between 1^2 and 2^2, $1 \in A$ and $2 \in B$.

$A \cup B = \mathbb{Q}$: Given $r \in \mathbb{Q}$, we can find $r \times r$ and compare it to 2. If it is less, it belongs to A. Otherwise, it belongs to B.

$a \in A, b \in B$ implies $a < b$: A consists of all rationals whose square is less than 2 (and all the negative rationals, but this case implies that $a < b$ because all elements of B are positive), and B consists of rationals whose square is greater than or equal to 2. So $a^2 < 2 < b^2$. Since the function $f(x) = x^2$ is increasing for $x > 0$, we get that $a < b$.

A contains no greatest element: Given $r \in A$, we need to show that there exists $r' \in A$ such that $r'^2 > r^2$. To do this, let $r' = r + \epsilon$ for $\epsilon > 0$. We need to choose rational ϵ such that $(r + \epsilon)^2 < 2$. Expanding and rearranging, we get $2r\epsilon + \epsilon^2 < 2 - r^2$. It would suffice for both $2r\epsilon$ and ϵ^2 to be less than $\frac{2-r^2}{2}$.

If $2r\epsilon < \frac{2-r^2}{2}$, then $\epsilon < \frac{2-r^2}{4r}$. And we need $\epsilon^2 < \frac{2-r^2}{2}$ to be true as well. If we assume the first statement to be true, then $\epsilon^2 < \frac{4+r^2-4r^4}{16r^2}$. Comparing that with $\frac{2-r^2}{2}$ and simplifying the inequality, we see that $\frac{4+r^2-4r^4}{16r^2} < \frac{2-r^2}{2}$ if $r \in \left(\sqrt{\frac{15-\sqrt{161}}{8}}, \sqrt{\frac{15+\sqrt{161}}{8}}\right)$. Since $\frac{15-\sqrt{161}}{8} < 2$ and $\frac{15+\sqrt{161}}{8} > 2$, r is in this range. This means, if we have $r \in A$ sufficiently close to x, then we can choose $\epsilon > 0$ such that $(r + \epsilon)^2 < 2$.

8. By the definition of a Dedekind cut, all elements of A are less than x.

Consider a real number $y < x$. If y is rational, then by definition it is an element of A, and not an upper bound. If it's irrational, then we can find a rational number between y and x. If we consider the decimal expansions of x and y and see where they differ, we could create a rational number by copying the digits of y up to that point and then terminating the number there. So y is less than some elements of A, and therefore not an upper bound.

9. Define the set $T : \{-s : s \in S\}$. By the least upper bound property, T has a least upper bound, $\sup(T)$. Since $\sup(T) \geq -s$ for all $s \in S$, we get $-\sup(T) \leq s$. Therefore, $-\sup(T)$ is a lower bound for S. Assume S has a lower bound, x, greater than $-\sup(T)$. Then $-\sup(T) < x < s$, so $-s < -x < \sup(T)$. This means $-x$ is an upper bound of T and it is less than the least upper bound of T, which is clearly impossible. Therefore, $-\sup(T)$ is the greatest upper bound of S.